(Study Scheme - English)

Programme Title
Electrical and Computer Engineering

Study Scheme

Applicable to students admitted in 2025-26 and thereafter

Major Programme Requirement

Studies in Electrical and Computer Engineering are divided into THREE main streams and students are required to specialize in one of the streams:
(1) Electronic Engineering;
(2) Computer Engineering; and
(3) Microelectronics Science and Engineering.

Electronic Engineering Stream

Students are required to complete a minimum of 72 units of courses as follows:

		Units
1.	School Package:	
	CHM1001, CSC1005, 1006, MAT1001, 1002, 2040, PHY1001, STA2001	22
2.	Required Courses:	
	CSC3100, ECE1811, 2001, 2050, 2811, 3001, 3080, 3810, 4010, ERG4901	26
3.	Elective Courses:	
(a)	18 units of courses selected from the following: Microelectronics and Devices Group: ECE3201, 3202, 4102, 4103, 4213, PHY3006, 3950 Signal Processing and Multimedia Group: ECE3510, 4512, 4513, 4520 Communications Group: ECE3050, 4002, 4003 Networking Group: CSC4160, ECE3280, 4006, 4007, 4016, 4050 AI Group: CSC3180, DDA3020, ECE3300, 4320 Computers Group: CSC3001, 3002, 3050, 3150, ECE4152, 4300, 4701, FTE4312 Robotics Group: AIR6021, 6023, 6064, 6205, 6206, 6207, 6211, 6212, ECE3040, 3060, 3250, 4011	18

(b)	6 units of courses selected from the following:	6
	Math and Physics Foundation Group: MAT2002, 3007, 3350, 4006, 4007, PHY1002, 2001, 3002, 3410, 4002, 4003, 4221	
	Interdisciplinary Extension and Research Group: BIM3001, BIO1001, ECO2011, ENE4008, ERG4902	
	Total:	**72**

Computer Engineering Stream

Students are required to complete a minimum of 72 units of courses as follows:

		Units
1.	School Package:	
	CHM1001, CSC1005, 1006, MAT1001, 1002, 2040, PHY1001, STA2001	22
2.	Required Courses:	
	CSC3002, 3100, ECE2050, 2810, 3070, 3080, 3810, 4010, 4016, 4300, ERG4901	29
3.	Elective Courses:	
(a)	15 units of courses selected from the following: Microelectronic and Devices Group: ECE3201, 3202, 4102, 4103	15
	Signal Processing and Multimedia Group: CSC3185, ECE3001, 3510, 4512, 4513, 4520	
	Communications Group: ECE3050, 4002, 4003, 4020	
	Networking Group: CSC4160, 4170, ECE3280, 4006, 4007, 4050	
	AI Group: CSC3180, 4008, DDA3020	
	Computers Group: CSC3001, 3050, 3150, 3170, 4001, 4005, 4120, 4150, 4180, 4190, DME3006, ECE3200, 4152, FTE4312	
	Robotics Group: AIR6021, 6023, 6064, 6202, 6205, 6206, 6207, 6211, 6212, ECE3040, 3060, 4011, 4310	
(b)	6 units of courses selected from the following:	6
	Math and Physics Foundation Group: MAT3007, 4006, 4007, PHY2001, 4002, 4003	

Interdisciplinary Extension and Research Group: BIM3001, BIO1001, ECO2011, ERG4902

Total: **72**

Microelectronics Science and Engineering Stream

Students are required to complete a minimum of 72 units of courses as follows:

		Units
1.	School Package:	
	CHM1001, CSC1005, 1006, MAT1001, 1002, 2040, PHY1001, STA2001	22
2.	Required Courses:	
	ECE1811, 2001, 2050, 2811, 3001, 3080, 3201, 3810, ERG4901, ISE3015, PHY3002, 3006	32
3.	Elective Courses:	
(a)	15 units of courses selected from the following:	15
	Computers Group: CSC3002, 3050, 3100	
	Integrated Circuits Group: ECE3202, 3510, 4014, 4102, 4103, 4213, 4320, 4701	
	Math Foundation Group: MAT2002, 3350	
(b)	3 units of course selected from the following:	3
	Application and Research Group: ECE4105, ERG4902, MAT4001, PHY2001, 2002, 2610, 3007, 3810, 4001, 4002, 4221	

Total: **72**

Explanatory Note:

[a] Students can choose to take either MAT2040 or MAT2042 to fulfill the graduation requirement.

[b] Students can choose to take either STA2001 or STA2001H to fulfill the graduation requirement.

[c] Students can choose to take either CSC1001 and CSC1002 or CSC1003 and CSC1004 or CSC1005 and CSC1006 to fulfill the graduation requirement.

[d] Students can choose to take either CHM1001 or CHM1011 to fulfill the graduation requirement.

[e] Students can choose to take either PHY1001 or PHY1011 to fulfill the graduation requirement.

[f] Courses included in the calculation of Major GPA for honors classification:

a) Electronic Engineering Stream

All Major Required and Major Elective courses at 2000 level and above.

b) Computer Engineering Stream

CSC1001, CSC1002, CSC1003, CSC1004, CSC1005, CSC1006 and all Major Required and Major Elective courses at 2000 level and above.

c) Microelectronics Science and Engineering Stream

All Major Required and Major Elective courses at 2000 level and above.

[g] Undergraduate students are allowed to take graduate courses offered under TPG programmes for credit if (a) they are Year 3 or above, their cumulative GPA is 3.00 or above and the application is approved by the course offering unit; or (b) by special approval of the Dean of the Graduate School, following recommendation of student's major School and the course offering unit. The postgraduate courses in the Major Elective course list will be included in the calculation of cumulative GPA and Major GPA.

課程名稱

電子與計算機工程

修讀辦法

二〇二五至二六年度及以後入學學生適用

主修課程要求

本課程分設三項專修範圍，學生須選修其中一項專修範圍：
(1) 電子工程；
(2) 計算機工程；及
(3) 微電子科學與工程

電子工程專修範圍

學生須至少修畢以下科目共 72 學分：

		學分
1.	學院課程：	
	CHM1001, CSC1005, 1006, MAT1001, 1002, 2040, PHY1001, STA2001	22
2.	必修科目：	
	CSC3100, ECE1811, 2001, 2050, 2811, 3001, 3080, 3810, 4010, ERG4901	26
3.	選修科目：	
(a)	從以下選修 18 學分： 微電子與器件组: ECE3201, 3202, 4102, 4103, 4213, PHY3006, 3950 信號處理與多媒體组: ECE3510, 4512, 4513, 4520 通信组: ECE3050, 4002, 4003 網路组: CSC4160, ECE3280, 4006, 4007, 4016, 4050 人工智能组: CSC3180, DDA3020, ECE3300, 4320 電腦组: CSC3001, 3002, 3050, 3150, ECE4152, 4300, 4701, FTE4312	18

	機器人组: AIR6021, 6023, 6064, 6205, 6206, 6207, 6211, 6212, ECE3040, 3060, 3250, 4011,	
(b)	從以下選修 6 學分：	6
	數學與物理基礎组: MAT2002, 3007, 3350, 4006, 4007, PHY1002, 2001, 3002, 3410, 4002, 4003, 4221	
	跨學科擴展组和研究组 :BIM3001, BIO1001, ECO2011, ENE4008, ERG4902	
	共：	**72**

計算機工程專修範圍

學生須至少修畢以下科目共 72 學分：

		學分
1.	學院課程：	
	CHM1001, CSC1005, 1006, MAT1001, 1002, 2040, PHY1001, STA2001	22
2.	必修科目：	
	CSC3002, 3100, ECE2050, 2810, 3070, 3080, 3810, 4010, 4016, 4300, ERG4901	29
3.	選修科目：	
(a)	從以下選修 15 學分： 微電子與器件组: ECE3201, 3202, 4102, 4103	15
	信號處理與多媒體组: CSC3185, ECE3001, 3510, 4512, 4513, 4520	
	通信组: ECE3050, 4002, 4003, 4020	
	網路组: CSC4160, 4170, ECE3280, 4006, 4007, 4050	
	人工智能组: CSC3180, 4008, DDA3020	
	電腦组: CSC3001, 3050, 3150, 3170, 4001, 4005, 4120, 4150, 4180, 4190, DME3006, ECE3200, 4152, FTE4312	
	機器人组: AIR6021, 6023, 6064, 6202, 6205, 6206, 6207, 6211, 6212, ECE3040, 3060, 4011, 4310	
(b)	從以下選修 6 學分：	6

數學與物理基礎组: MAT3007, 4006, 4007, PHY2001, 4002, 4003

跨學科擴展和研究组: BIM3001, BIO1001, ECO2011, ERG4902

共： **72**

微電子科學與工程專修範圍

學生須至少修畢以下科目共 72 學分：

		學分
1.	學院課程：	
	CHM1001, CSC1005, 1006, MAT1001, 1002, 2040, PHY1001, STA2001	22
2.	必修科目：	
	ECE1811, 2001, 2050, 2811, 3001, 3080, 3201, 3810, ERG4901, ISE3015, PHY3002, 3006	32
3.	選修科目：	
(a)	從以下選修 15 學分：	15
	電腦组: CSC3002, 3050, 3100	
	集成電路組: ECE3202, 3510, 4014, 4102, 4103, 4213, 4320, 4701	
	數學基礎组: MAT2002, 3350	
(b)	從以下選修 3 學分： 應用和研究组: ECE4105, ERG4902, MAT4001, PHY2001, 2002, 2610, 3007, 3810, 4001, 4002, 4221	3

共： **72**

註：

[a] 學生可以修讀 MAT2040 或 MAT2042 以滿足畢業要求。

[b] 學生可以修讀 STA2001 或 STA2001H 以滿足畢業要求。

[c] 學生可以修讀 CSC1001, CSC1002 或 CSC1003, CSC1004 或 CSC1005, CSC1006 以滿足畢業要求。

[d] 學生可以修讀 CHM1001 或 CHM1011 以滿足畢業要求。

[e] 學生可以修讀 PHY1001 或 PHY1011 以滿足畢業要求。

[f] 下列科目將會計入主修科目之平均績點，並用以釐定學位等級：

a) 電子工程專修範圍

所有 2000 及以上程度的主修必修和主修选修科目

b) 計算機工程專修範圍

CSC1001, CSC1002, CSC1003, CSC1004, CSC1005, CSC1006, 以及所有 2000 及以上程度的主修必修和主修选修科目

c) 微電子科學與工程專修範圍

所有 2000 及以上程度的主修必修和主修选修科目

[g] 本科學生可申請修讀本修讀辦法中的授課型碩士項目科目，申請須滿足(a)學生已至大三或以上年級，累計平均積點為 3.00 或以上，並獲開課學院批准；或者（b）經其所屬學院及開課學院推薦，獲得研究生院院長的特別批准。專業選修中的研究生科目將會計入累計平均積點及主修科目之平均績點。

(Recommended Course Pattern - English)

<u>Recommended Course Pattern</u>

1. Sufficient units should be allowed in each term for students to fulfill the University Core Requirements, which include: (i) 3 units of Chinese; (ii) 12 units of English; (iii) 1 unit of IT; (iv) 18 units of General Education; and (v) 2 units of Physical Education and Health.
2. Programmes with different streams/concentrations are required to provide the recommended pattern for each stream/concentration.

	Major Programme Requirement of <u>Electronic Engineering Stream</u>	
	Recommended Course Pattern	**Units**
First Year of Attendance	1st term School Package: CHM1001(or CHM1011), CSC1005(or CSC1001 or CSC1003), MAT1001	9
	2nd term School Package: CSC1006(or CSC1002 or CSC1004), MAT1002, MAT2040(or MAT2042), PHY1001(or PHY1011)	10
Second Year of Attendance	1st term School Package: STA2001(or STA2001H) Major Required: ECE1811, 2001	 3 5
	2nd term Major Required: ECE2050, 2811, 3001	8
Third Year of Attendance	1st term Major Required: ECE3080, 3810 Major Elective(s): Any two elective courses	 4 6
	2nd term Major Required: ECE4010 Major Elective(s): Any three elective courses	 3 9
Fourth Year of Attendance	1st term Major Required: CSC3100, ERG4901 Major Elective(s): Any one elective course	 6 3
	2nd term Major Elective(s): Any two elective courses	6
	Total (Major Requirement including School Package):	**72**

(Recommended Course Pattern - Chinese)

修課推介

1. 每學期均須預留足夠學分讓同學符合大學核心課程要求，包括：（一）中文三學分；（二）英文十二學分；（三）信息科技一學分；（四）通識教育十八學分及（五）體育與健康兩學分。
2. 有不同專修範圍的課程須為每項專修範圍提供修課推介。

	電子工程主修課程要求	
	修課推介	**學分**
第一修業學年	第一學期 學院課程：CHM1001(or CHM1011), CSC1005(or CSC1001 or CSC1003), MAT1001	9
	第二學期 學院課程：CSC1006(or CSC1002 or CSC1004), MAT1002, MAT2040(or MAT2042), PHY1001(or PHY1011)	10
第二修業學年	第一學期 學院課程：STA2001(or STA2001H) 主修必修科目：ECE1811, 2001	3 5
	第二學期 主修必修科目：ECE2050, 2811, 3001	8
第三修業學年	第一學期 主修必修科目：ECE3080, 3810 主修選修科目：两科選修科目	4 6
	第二學期 主修必修科目：ECE4010 主修選修科目：三科選修科目	3 9
第四修業學年	第一學期 主修必修科目：CSC3100, ERG4901 主修選修科目：一科選修科目	6 3
	第二學期 主修選修科目：兩科選修科目	6
	總計（主修要求包括學院課程）	**72**

(Recommended Course Pattern - English)

Recommended Course Pattern

1. Sufficient units should be allowed in each term for students to fulfill the University Core Requirements, which include: (i) 3 units of Chinese; (ii) 12 units of English; (iii) 1 unit of IT; (iv) 18 units of General Education; and (v) 2 units of Physical Education and Health.
2. Programmes with different streams/concentrations are required to provide the recommended pattern for each stream/concentration.

	Major Programme Requirement of <u>Computer Engineering Stream</u>	
	Recommended Course Pattern	**Units**
First Year of Attendance	1st term School Package: CHM1001(or CHM1011), CSC1005(or CSC1001 or CSC1003), MAT1001	9
	2nd term School Package: CSC1006(or CSC1002 or CSC1004), MAT1002, MAT2040(or MAT2042), PHY1001(or PHY1011)	10
Second Year of Attendance	1st term School Package: STA2001(or STA2001H) Major Required: CSC3002, ECE2050, 2810	3 7
	2nd term Major Required: CSC3100, ECE3070	6
Third Year of Attendance	1st term Major Required: ECE3080, 3810, 4016 Major Elective(s): Any one elective course	7 3
	2nd term Major Required: ECE4010 Major Elective(s): Any three elective courses	3 9
Fourth Year of Attendance	1st term Major Required: ECE4300, ERG4901 Major Elective(s): Any one elective course	6 3
	2nd term Major Elective(s): Any two elective courses	6
	Total (Major Requirement including School Package):	**72**

(Recommended Course Pattern - Chinese)

<u>修課推介</u>

1. 每學期均須預留足夠學分讓同學符合大學核心課程要求，包括：（一）中文三學分；（二）英文十二學分；（三）信息科技一學分；（四）通識教育十八學分及（五）體育與健康兩學分。
2. 有不同專修範圍的課程須為每項專修範圍提供修課推介。

<table>
<tr><td></td><td><u>計算機工程</u>主修課程要求</td><td></td></tr>
<tr><td></td><td>修課推介</td><td>學分</td></tr>
<tr><td rowspan="2">第一修業學年</td><td>第一學期
學院課程：CHM1001(or CHM1011), CSC1005(or CSC1001 or CSC1003), MAT1001</td><td>9</td></tr>
<tr><td>第二學期
學院課程：CSC1006(or CSC1002 or CSC1004), MAT1002, MAT2040(or MAT2042), PHY1001(or PHY1011)</td><td>10</td></tr>
<tr><td rowspan="2">第二修業學年</td><td>第一學期
學院課程：STA2001(or STA2001H)
主修必修科目：CSC3002, ECE2050, 2810</td><td>3
7</td></tr>
<tr><td>第二學期
主修必修科目：CSC3100, ECE3070</td><td>6</td></tr>
<tr><td rowspan="2">第三修業學年</td><td>第一學期
主修必修科目： ECE3080, 3810, 4016
主修選修科目：一科選修科目</td><td>7
3</td></tr>
<tr><td>第二學期
主修必修科目：ECE4010
主修選修科目：三科選修科目</td><td>3
9</td></tr>
<tr><td rowspan="2">第四修業學年</td><td>第一學期
主修必修科目：ECE4300, ERG4901
主修選修科目：一科選修科目</td><td>6
3</td></tr>
<tr><td>第二學期
主修選修科目：兩科選修科目</td><td>6</td></tr>
<tr><td colspan="2" align="right">總計（主修要求包括學院課程）</td><td>72</td></tr>
</table>

(Recommended Course Pattern - English)

<u>Recommended Course Pattern</u>

1. Sufficient units should be allowed in each term for students to fulfill the University Core Requirements, which include: (i) 3 units of Chinese; (ii) 12 units of English; (iii) 1 unit of IT; (iv) 18 units of General Education; and (v) 2 units of Physical Education and Health.
2. Programmes with different streams/concentrations are required to provide the recommended pattern for each stream/concentration.

	Major Programme Requirement of <u>Microelectronics Science and Engineering Stream</u>	
	Recommended Course Pattern	**Units**
First Year of Attendance	1st term School Package: CHM1001(or CHM1011), CSC1005(or CSC1001 or CSC1003), MAT1001	9
	2nd term School Package: CSC1006(or CSC1002 or CSC1004), MAT1002, MAT2040(or MAT2042), PHY1001(or PHY1011)	10
Second Year of Attendance	1st term School Package: STA2001(or STA2001H) Major Required: ECE1811, 2001, PHY3002	3 8
	2nd term Major Required: ECE2050, 2811, 3001	8
Third Year of Attendance	1st term Major Required: ECE3080, 3201, 3810 Major Elective(s): Any two elective courses	7 6
	2nd term Major Required: ISE3015, PHY3006 Major Elective(s): Any one elective course	6 3
Fourth Year of Attendance	1st term Major Required: ERG4901 Major Elective(s): Any one elective course	3 3
	2nd term Major Elective(s): Any two elective courses	6
	Total (Major Requirement including School Package):	**72**

(Recommended Course Pattern - Chinese)

修課推介

1. 每學期均須預留足夠學分讓同學符合大學核心課程要求，包括：（一）中文三學分；（二）英文十二學分；（三）信息科技一學分；（四）通識教育十八學分及（五）體育與健康兩學分。
2. 有不同專修範圍的課程須為每項專修範圍提供修課推介。

	微電子科學與工程主修課程要求	
	修課推介	**學分**
第一修業學年	第一學期 學院課程：CHM1001(or CHM1011), CSC1005(or CSC1001 or CSC1003), MAT1001	9
	第二學期 學院課程：CSC1006(or CSC1002 or CSC1004), MAT1002, MAT2040(or MAT2042), PHY1001(or PHY1011)	10
第二修業學年	第一學期 學院課程：STA2001(or STA2001H) 主修必修科目：ECE1811, ECE2001, PHY3002	3 8
	第二學期 主修必修科目：ECE2050, 2811, 3001	8
第三修業學年	第一學期 主修必修科目：ECE3080, 3201, 3810 主修選修科目：兩科選修科目	7 6
	第二學期 主修必修科目：ISE3015, PHY3006 主修選修科目：一科選修科目	6 3
第四修業學年	第一學期 主修必修科目：ERG4901 主修選修科目：一科選修科目	3 3
	第二學期 主修選修科目：兩科選修科目	6
	總計（主修要求包括學院課程）	**72**

Course List

I. School Package for the School of Science and Engineering

Course Code		Course Title (English)	Course Title (Chinese)	Unit(s)
Option J#	CHM1001	General Chemistry	普通化學	3
Option K#	CHM1011	Honours General Chemistry	普通化學榮譽課程	3
Option C#	CSC1001	Introduction to Computer Science: Programming Methodology	計算機科學導論：程序設計方法	3
	CSC1002	Computational Laboratory	計算機實驗	1
Option D#	CSC1003	Introduction to Computer Science and Java Programming	計算機科學與 Java 程序設計導論	3
	CSC1004	Computational Laboratory Using Java	Java 程序設計實驗	1
Option E#	CSC1005	Introduction to Computer Engineering: Programming and Applications	計算機工程導論：程序設計与应用	3
	CSC1006	Artificial Intelligence for Science and Engineering	人工智能在科學與工程中的應用	1
MAT1001		Calculus I	微積分（一）	3
MAT1002		Calculus II	微積分（二）	3
Option A#	MAT2040	Linear Algebra	線性代數	3
Option B#	MAT2042	Honours Linear Algebra	線性代數榮譽課程	3
Option H#	PHY1001	Mechanics	力學	3
Option I#	PHY1011	Honours Mechanics	力學榮譽課程	3
Option F#	STA2001	Probability and Statistics I	概率及統計（一）	3
Option G#	STA2001H	Honours Probability and Statistics I	概率及統計榮譽課程（一）	3

Remark:
1. Students can choose to take either Option A or Option B to fulfill the graduation requirement.
2. Students can choose to take either Option C or Option D or Option E to fulfill the graduation requirement.
3. Students can choose to take either Option F or Option G to fulfill the graduation requirement.
4. Students can choose to take either Option H or Option I to fulfill the graduation requirement.
5. Students can choose to take either Option J or Option K to fulfill the graduation requirement.

II. Major Required courses

Electronic Engineering Stream

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)
CSC3100	Data Structures	數據結構	3
ECE1811	Honours Electronic Circuit Design Laboratory	電子線路設計實驗榮譽課程	2
ECE2001	Basic Circuit Theory	基本電路理論	3
ECE2050	Digital Logic and Systems	數字邏輯與系統	3

ECE2811	Honours Digital Systems Design Laboratory	數字系統設計實驗榮譽課程	2
ECE3001	Signals and Systems	信號與系統	3
ECE3080	Introduction to Embedded Systems	嵌入式系統概論	3
ECE3810	Embedded Systems Laboratory	嵌入式系統實驗	1
ECE4010	Machine Intelligence and Applications	機器學習及應用	3
ERG4901	Capstone Project I	畢業設計（一）	3

Computer Engineering Stream

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)
CSC3002	C/C++ Programming	C/C++程序設計	3
CSC3100	Data Structures	數據結構	3
ECE2050	Digital Logic and Systems	數字邏輯與系統	3
ECE2810	Digital Systems Design Laboratory	數字系統設計實驗	1
ECE3070	Computer Organization and Systems	計算機組成原理與系統	3
ECE3080	Introduction to Embedded Systems	嵌入式系統概論	3
ECE3810	Embedded Systems Laboratory	嵌入式系統實驗	1
ECE4010	Machine Intelligence and Applications	機器學習及應用	3
ECE4016	Computer Networks	計算機網絡	3
ECE4300	Systems Design in Human-Computer Interaction	人機交互系統設計	3
ERG4901	Capstone Project I	畢業設計（一）	3

Microelectronics Science and Engineering Stream

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)
ECE1811	Honours Electronic Circuit Design Laboratory	電子線路設計實驗榮譽課程	2
ECE2001	Basic Circuit Theory	基本電路理論	3
ECE2050	Digital Logic and Systems	數字邏輯與系統	3
ECE2811	Honours Digital Systems Design Laboratory	數字系統設計實驗榮譽課程	2
ECE3001	Signals and Systems	信號與系統	3
ECE3080	Introduction to Embedded Systems	嵌入式系統概論	3
ECE3201	Introduction to Microelectronic Circuits	微電子電路導論	3
ECE3810	Embedded Systems Laboratory	嵌入式系統實驗	1
ERG4901	Capstone Project I	畢業設計（一）	3
ISE3015	Microfabrication Technology	微加工技術	3
PHY3002	Electrodynamics I	電動力學（一）	3
PHY3006	Physics of Semiconductors Devices	半導體器件物理	3

III. Elective courses

Electronic Engineering Stream

Microelectronics and Devices Group:

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)
ECE3201	Introduction to Microelectronic Circuits	微電子電路導論	3
ECE3202	Analog Integrated Circuits	模擬集成電路	3
ECE4102	CMOS Digital Integrated Circuits Design	CMOS 數字集成電路設計	3
ECE4103	Very Large Scale Integrated Circuit Design Automation	超大規模集成電路設計自動化	3
ECE4213	Radio Frequency Integrated Circuits	射頻集成電路	3
PHY3006	Physics of Semiconductors Devices	半導體器件物理	3
PHY3950	Basic Electronics	電子學基礎	3

Signal Processing and Multimedia Group:

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)
ECE3510	Digital Signal Processing	數字信號處理	3
ECE4512	Digital Image Processing	數字圖像處理	3
ECE4513	Image Processing and Computer Vision	圖像處理和計算機視覺	3
ECE4520	3D Data Processing	三維數據處理	3

Communications Group:

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)
ECE3050	Principles of Communication Systems	通信系統原理	3
ECE4002	Digital Communications	數字通信	3
ECE4003	Wireless Communication Systems	無線通信系統	3

Networking Group:

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)
CSC4160	Cloud Computing	雲端計算	3
ECE3280	Networks: Technology, Economics and Society	網絡：技術、經濟和社會	3
ECE4006*	Applied Stochastic Processes	應用隨機過程	3
ECE4007	Computer and Network Security	計算機及網絡安全	3
ECE4016	Computer Networks	計算機網絡	3
ECE4050	Principles and Practices of Communication Networks	通信網絡原理與實踐	3

AI Group:

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)
CSC3180	Fundamentals of Artificial Intelligence	人工智能之基本原理	3
DDA3020	Machine Learning	機器學習	3
ECE3300	Introduction to AI Infrastructure	人工智能基礎架構導論	3
ECE4320	Hardware Acceleration for Efficient AI Computing	硬件加速的高效人工智能計算	3

Computers Group:

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)
CSC3001	Discrete Mathematics	離散數學	3
CSC3002	C/C++ Programming	C/C++程序設計	3
CSC3050	Computer Architecture	計算機體系結構	3
CSC3150	Operating System	操作系統	3
ECE4152	Mobile Application Design & Implementation	移動端應用程序設計與實現	3
ECE4300	Systems Design in Human-Computer Interaction	人機交互系統設計	3
ECE4701	Advanced Embedded Systems	高級嵌入式系統	3
FTE4312	Blockchain and Decentralized Applications	區塊鏈與去中心化應用	3

Robotics Group:

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)
AIR6021	Robot Manipulation	機器人操作	3
AIR6023	Math Fundamentals for Robotics	機器人數學基礎	3
AIR6064	Human-Machine Interaction	人機交互	3
AIR6205	Robot Design and Manufacturing	機器人設計與製造	3
AIR6206	Mobile Robotics	移動機器人	3
AIR6207	Micro-Nanorobotics	微納米機器人	3
AIR6211	Soft Robotics	軟體機器人	3
AIR6212	Introduction to Haptics	機器人觸覺導論	3
ECE3040	Introduction to Linear Systems	線性系統導論	3
ECE3060	Introduction to Robotics	機械人學概論	3
ECE3250	System & Control	系統與控制	3
ECE4011	Robotics and Intelligent Systems	機械人與智能系統	3

Math and Physics Foundation Group:

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)
MAT2002	Ordinary Differential Equations	常微分方程	3
MAT3007	Optimization	最優化	3

MAT3350	Introduction to Information Theory	信息論導論	3
MAT4006	Introduction to Coding Theory	編碼理論導論	3
MAT4007	Introduction to Cryptography	密碼學導論	3
PHY1002	Physics Laboratory	物理實驗	2
PHY2001	Electricity and Magnetism	電磁學	3
PHY3002	Electrodynamics I	電動力學（一）	3
PHY3410	Quantum Mechanics and its Applications I	量子力學（一）	3
PHY4002	Electrodynamics II	電動力學（二）	3
PHY4003	Introduction to Photonic Crystals	光子晶體簡介	3
PHY4221	Quantum Mechanics	量子力學	3

Interdisciplinary Extension and Research Group:

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)
BIM3001	Bioinformatics	生物信息學	3
BIO1001	General Biology	普通生物學	3
ECO2011	Basic Microeconomics	微觀經濟學基礎	3
ENE4008	Power Electronics	電力電子	3
ERG4902	Capstone Project II	畢業設計（二）	3

Computer Engineering Stream

Microelectronic and Devices Group:

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)
ECE3201	Introduction to Microelectronic Circuits	微電子電路導論	3
ECE3202	Analog Integrated Circuits	模擬集成電路	3
ECE4102	CMOS Digital Integrated Circuits Design	CMOS 數字集成電路設計	3
ECE4103	Very Large Scale Integrated Circuit Design Automation	超大規模集成電路設計自動化	3

Signal Processing and Multimedia Group:

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)
CSC3185	Fundamentals of Artificial Intelligence	多媒體系統導論	3
ECE3001	Signals and Systems	信號與系統	3
ECE3510	Digital Signal Processing	數字信號處理	3
ECE4512	Digital Image Processing	數字圖像處理	3
ECE4513	Image Processing and Computer Vision	圖像處理和計算機視覺	3
ECE4520	3D Data Processing	三維數據處理	3

Communications Group:

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)
ECE3050	Principles of Communication Systems	通信系統原理	3
ECE4002	Digital Communications	數字通信	3
ECE4003	Wireless Communication Systems	無線通信系統	3
ECE4020	Telecommunication Switching and Network Systems	電信交換與網絡系統	3

Networking Group:

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)
CSC4160	Cloud Computing	雲端計算	3
CSC4170	Social Networks	社交網絡	3
ECE3280	Networks: Technology, Economics and Society	網絡：技術、經濟和社會	3
ECE4006*	Applied Stochastic Processes	應用隨機過程	3
ECE4007	Computer and Network Security	計算機及網絡安全	3
ECE4050	Principles and Practices of Communication Networks	通信網絡原理與實踐	3

AI Group:

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)
CSC3180	Fundamentals of Artificial Intelligence	人工智能之基本原理	3
CSC4008	Techniques for Data Mining	數據挖掘技術	3
DDA3020	Machine Learning	機器學習	3

Computers Group:

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)
CSC3001	Discrete Mathematics	離散數學	3
CSC3050	Computer Architecture	計算機體系結構	3
CSC3150	Operating System	操作系統	3
CSC3170	Database System	數據庫系統	3
CSC4001	Software Engineering	軟件工程	3
CSC4005	Parallel Programming	並行程序設計	3
CSC4120	Design and Analysis of Algorithms	算法設計及分析	3
CSC4150	Mobile Computing with Internet of Things	物聯網移動計算	3
CSC4180	Compiler Construction	編譯器設計	3
CSC4190	Internet Programming and Applications	互聯網編程與應用	3
DME3006	Computer Aided Design	計算機輔助設計	3
ECE3200	Video Games Design and Development	電子遊戲設計與開發	3

ECE4152	Mobile Application Design & Implementation	移動端應用程序設計與實現	3
FTE4312	Blockchain and Decentralized Applications	區塊鏈與去中心化應用	3

Robotics Group:

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)
AIR6021	Robot Manipulation	機器人操作	3
AIR6023	Math Fundamentals for Robotics	機器人數學基礎	3
AIR6064	Human-Machine Interaction	人機交互	3
AIR6202	System and Control	系統與控制	3
AIR6205	Robot Design and Manufacturing	機器人設計與製造	3
AIR6206	Mobile Robotics	移動機器人	3
AIR6207	Micro-Nanorobotics	微納米機器人	3
AIR6211	Soft Robotics	軟體機器人	3
AIR6212	Introduction to Haptics	機器人觸覺導論	3
ECE3040	Introduction to Linear Systems	線性系統導論	3
ECE3060	Introduction to Robotics	機械人學概論	3
ECE4011	Robotics and Intelligent Systems	機械人與智能系統	3
ECE4310	Programming for Robotics	機器人操作系統與編程	3

Math and Physics Foundation Group:

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)
MAT3007	Optimization	最優化	3
MAT4006	Introduction to Coding Theory	編碼理論導論	3
MAT4007	Introduction to Cryptography	密碼學導論	3
PHY2001	Electricity and Magnetism	電磁學	3
PHY4002	Electrodynamics II	電動力學（二）	3
PHY4003	Introduction to Photonic Crystals	光子晶體簡介	3

Interdisciplinary Extension and Research Group:

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)
BIM3001	Bioinformatics	生物信息學	3
BIO1001	General Biology	普通生物學	3
ECO2011	Basic Microeconomics	微觀經濟學基礎	3
ERG4902	Capstone Project II	畢業設計（二）	3

Microelectronics Science and Engineering Stream

Computers Group:

Course Code	Course Title (English)	Course Title (Chinese)	Unit(s)

CSC3002	C/C++ Programming	C/C++程序設計	3
CSC3050	Computer Architecture	計算機體系結構	3
CSC3100	Data Structures	數據結構	3

Integrated Circuits Group:

Course Code	**Course Title (English)**	**Course Title (Chinese)**	**Unit(s)**
ECE3202	Analog Integrated Circuits	模擬集成電路	3
ECE3510	Digital Signal Processing	數字信號處理	3
ECE4014	Power Electronics Design	功率電子學設計	3
ECE4102	CMOS Digital Integrated Circuits Design	CMOS 數字集成電路設計	3
ECE4103	Very Large Scale Integrated Circuit Design Automation	超大規模集成電路設計自動化	3
ECE4213	Radio Frequency Integrated Circuits	射頻集成電路	3
ECE4320	Hardware Acceleration for Efficient AI Computing	硬件加速的高效人工智能計算	3
ECE4701	Advanced Embedded Systems	高級嵌入式系統	3

Math Foundation Group:

Course Code	**Course Title (English)**	**Course Title (Chinese)**	**Unit(s)**
MAT2002	Ordinary Differential Equations	常微分方程	3
MAT3350	Introduction to Information Theory	信息論導論	3

Application and Research Group:

Course Code	**Course Title (English)**	**Course Title (Chinese)**	**Unit(s)**
ECE4105	Introduction to Microelectromechanical Systems (MEMS)	微機電系統（MEMS）導論	3
ERG4902	Capstone Project II	畢業設計（二）	3
MAT4001	Numerical Analysis	數值分析	3
PHY2001	Electricity and Magnetism	電磁學	3
PHY2002	Thermodynamics	熱力學	3
PHY2610	Mathematical Methods in Physics I	數學物理方法（一）	3
PHY3007	Optoelectronics	光電子學	3
PHY3810	Modern Optical Physics	現代光學物理	3
PHY4001	Solid-State Physics	固體物理	3
PHY4002	Electrodynamics II	電動力學（二）	3
PHY4221	Quantum Mechanics	量子力學	3

* The course title of ECE4006 is Performance Evaluation of Communication Networks for students who study the course before 2025-26 term 1. For students who study this course from 2025-26 term 1, the course title is Applied Stochastic Processes.

在 2025-26 學年第一学期前修讀 ECE4006，科目名稱為「通信網絡性能分析」。在 2025-26 學年第一学期及以後修讀該科目，科目名稱為「應用隨機過程」。